Stone I and Stone II three-phase relative permeability

Step-by-step equations, matched to the Python variables used in the script

0. Inputs and the saturation grid

Symbol	Python	Value	Meaning
S_{wr}	`Swr`	0.10	irreducible (residual) water saturation
S_{gr}	`Sgr`	0.05	critical (residual) gas saturation: gas flows only above it
$S_{or,ow}$	`Sor_ow`	0.40	residual oil in the oil-water system
$S_{or,og}$	`Sor_og`	0.20	residual oil in the gas-oil system
$k_{ro,ow}^{0}$, $k_{ro,og}^{0}$	`kro_ow, kro_og`	0.8, 0.8	oil end points (Stone I assumes they are equal)
k_{rw}^{0}, k_{rg}^{0}	`krw0, krg0`	0.3, 0.3	water and gas end points
n_w, n_{ow}, n_g, n_{og}	`nw, now, ng, nog`	2, 2, 2, 2	Corey exponents

The three saturations always add up to one:

$$S_o = 1 - S_w - S_g \tag{1}$$

In Python build the grid with `np.meshgrid` (every Sw combined with every Sg), then keep only the points of the triangle with $S_w + S_g \le 1$. Two 1-D arrays subtracted elementwise give only a diagonal line, not a grid.

```
Sw_1d = np.linspace(Swr, 1.0, 101)
Sg_1d = np.linspace(0.0, 1.0 - Swr, 101)
SW, SG = np.meshgrid(Sw_1d, Sg_1d)
valid = SW + SG <= 1 + 1e-12
Sw, Sg = SW[valid], SG[valid]
So = np.clip(1 - Sw - Sg, 0, 1)
```

1. Normalized saturations for the two-phase curves

Both are clipped to [0, 1]. Without the clip, $(1 - S_d)^2$ grows again beyond the residual saturations.

$$S_{wd} = \frac{S_w - S_{wr}}{1 - S_{wr} - S_{or,ow}} \tag{2}$$

$$S_{gd} = \frac{S_g - S_{gr}}{1 - S_{wr} - S_{or,og} - S_{gr}} \tag{3}$$

2. Two-phase relative permeabilities (Corey)

Water and oil in the oil-water system depend only on S_w. Gas and oil in the gas-oil system depend only on S_g.

$$k_{rw} = k_{rw}^{0} S_{wd}^{n_w} \qquad k_{rg} = k_{rg}^{0} S_{gd}^{n_g} \tag{4}$$

$$k_{row} = k^0_{ro,\,ow}\,(1 - S_{wd})^{n_{ow}} \qquad k_{rog} = k^0_{ro,\,og}\,(1 - S_{gd})^{n_{og}} \tag{5}$$

When each curve is exactly zero (these are the grey zones in the ternary plots):

Curve	Zero when	Reaches its end point when
k_{rw}	$S_w \le S_{wr} = 0.10$	$S_w \ge 1 - S_{or,ow} = 0.60$
k_{rg}	$S_g \le S_{gr} = 0.05$	$S_g \ge 1 - S_{wr} - S_{or,og} = 0.70$
k_{row}	$S_w \ge 0.60$	$S_w \le S_{wr}$
k_{rog}	$S_g \ge 0.70$	$S_g \le S_{gr}$

3. Oil gate: normalized oil saturation S_{oD} (textbook Eq. 1.27)

S_{oD} uses the **fixed** residual $S_{or,ow}$ and subtracts S_{gr} in the denominator. Do **not** clip it: the three-way test below needs the raw value.

$$S_{oD} = \frac{S_o - S_{or,\,ow}}{1 - S_{wr} - S_{or,\,ow} - S_{gr}} \tag{6}$$

4. Stone I oil relative permeability

4a. Minimum oil saturation S_{om} (Fayers and Matthews). The weight a runs from 0 at $S_g = 0$ to 1 at $S_g = 1 - S_{wr} - S_{or,og}$; clip it to [0, 1].

$$a = \frac{S_g}{1 - S_{wr} - S_{or,\,og}} \tag{7}$$

$$S_{om} = (1 - a)\,S_{or,\,ow} + a\,S_{or,\,og} \tag{8}$$

4b. Three-phase normalized saturations. D is the common denominator.

$$D = 1 - S_{wr} - S_{om} \tag{9}$$

$$S^*_w = \frac{S_w - S_{wr}}{D} \qquad S^*_o = \frac{S_o - S_{om}}{D} \qquad S^*_g = \frac{S_g}{D} \tag{10}$$

$$S^*_w + S^*_o + S^*_g = 1 \tag{11}$$

Equation (11) is a good check to print in your code: it must equal 1 at every point.

4c. Stone I formula:

$$k^{I}_{ro} = \frac{S^*_o\,k_{row}\,k_{rog}}{k^0_{ro,\,ow}\,(1 - S^*_w)\,(1 - S^*_g)} \tag{12}$$

Because the three starred saturations add to one, the denominator equals $(S_o^* + S_g^*)(S_o^* + S_w^*)$. It is zero only at corner points where $S_o^* = 0$, and the numerator is zero there too (0/0). In code, use the formula only where $S_o^* > 0$ and put kro = 0 elsewhere; use a dummy denominator of 1 in the unused branch so no division by zero happens.

4d. Final oil relative permeability (textbook Eq. 1.26):

Condition	k_{ro}
$S_{oD} < 0$ ($S_o < S_{or,ow}$)	0
$0 \le S_{oD} \le 1$	k_{ro}^{I} from (4c), or 0 if $S_o^* \le 0$
$S_{oD} > 1$	$k_{ro,ow}^{0}$ (the end point)

Limits to test: at $S_g = 0$, $k_{ro} = k_{row}$ (outside the $S_{oD} > 1$ region); at $S_w = S_{wr}$, $k_{ro} = k_{rog}$ for the Stone I formula without the gate. The gate cuts k_{ro} to 0 below $S_o = S_{or,ow}$ even where S_{om} is lower, so the textbook version deviates from k_{rog} there.

5. Stone II oil relative permeability

No S_{om}, no starred saturations and no gate. It uses the two-phase curves and k_{rw}, k_{rg} directly:

$$k_{ro}^{II} = k_{ro}^{0}\left[\left(\frac{k_{row}}{k_{ro}^{0}} + k_{rw}\right)\left(\frac{k_{rog}}{k_{ro}^{0}} + k_{rg}\right) - \left(k_{rw} + k_{rg}\right)\right] \tag{13}$$

$$k_{ro} = \max\left(k_{ro}^{II},\, 0\right) \tag{14}$$

Limits: at $S_g = 0$ ($k_{rg} = 0$) it gives k_{row}; at $S_w = S_{wr}$ ($k_{rw} = 0$, $k_{row} = k_{ro}^{0}$) it gives k_{rog}. The raw value is negative at about half of the grid points, so the max(·, 0) is required. Residual oil is not enforced in the middle of the triangle; it enters only through k_{row} and k_{rog}.

6. Worked example to check your code

Point $S_w = 0.30$, $S_g = 0.20$, so $S_o = 0.50$. Every number below was computed by the script in section 7; your Python should print the same values.

Step	Quantity	Value
1	S_{wd}, S_{gd}	0.4000, 0.2308
2	k_{rw}, k_{rg}	0.0480, 0.0160
2	k_{row}, k_{rog}	0.2880, 0.4734
3	S_{oD} (in 0..1, so the Stone I formula is used)	0.2222
4a	a, S_{om}	0.2857, 0.3429
4b	D	0.5571
4b	S_w^*, S_o^*, S_g^* (sum = 1)	0.3590, 0.2821, 0.3590
4c	$(1 - S_w^*)(1 - S_g^*)$	0.4109
4d	**k_{ro} Stone I**	**0.1170**
5	Stone II raw value, then k_{ro}	0.1472 → **0.1472**

Four more check points:

S_w	S_g	S_o	k_{ro} Stone I	k_{ro} Stone II	note
0.10	0.40	0.50	0.1704	0.1704	= k_{rog} ($S_w = S_{wr}$)
0.30	0.00	0.70	0.2880	0.2880	= k_{row} ($S_g = 0$)
0.10	0.55	0.35	0.0000	0.0426	$S_o < 0.4$: Stone I gate gives 0
0.40	0.10	0.50	0.0757	0.0952	both positive

7. Complete Python (runs as is, needs only NumPy)

This is the exact code used to produce the table above; it also reproduces the saturation grid of section 0.

```
import numpy as np

# ---- inputs -------------------------------------------------------------
Swr, Sgr = 0.10, 0.05          # irreducible water, critical (residual) gas
Sor_ow, Sor_og = 0.40, 0.20    # residual oil: oil-water and gas-oil systems
kro_ow = kro_og = 0.80         # oil end points (Stone I assumes they are equal)
krw0, krg0 = 0.30, 0.30        # water and gas end points
nw = now = ng = nog = 2        # Corey exponents

clip01 = lambda x: np.clip(x, 0.0, 1.0)

def two_phase(Sw, Sg):
    """Step 1 and 2: normalized saturations and two-phase curves."""
    Swd = clip01((Sw - Swr) / (1 - Swr - Sor_ow))
    Sgd = clip01((Sg - Sgr) / (1 - Swr - Sor_og - Sgr))
    krw  = krw0 * Swd**nw
    krg  = krg0 * Sgd**ng
    krow = kro_ow * (1 - Swd)**now
    krog = kro_og * (1 - Sgd)**nog
    return krw, krg, krow, krog

def stone1(Sw, Sg):
    So = 1.0 - Sw - Sg
    krw, krg, krow, krog = two_phase(Sw, Sg)
    SoD = (So - Sor_ow) / (1 - Swr - Sor_ow - Sgr)        # step 3 (raw, NOT clipped)
    a   = clip01(Sg / (1 - Swr - Sor_og))                 # step 4
    Som = (1 - a) * Sor_ow + a * Sor_og
    D   = 1 - Swr - Som
    Sw_s, So_s, Sg_s = (Sw - Swr) / D, (So - Som) / D, Sg / D
    den   = (1 - Sw_s) * (1 - Sg_s)
    stone = np.where(So_s > 0, So_s * krow * krog / (kro_ow * np.where(So_s > 0, den, 1.0)), 0.0)
    kro = np.where(SoD < 0, 0.0, np.where(SoD > 1, kro_ow, stone))
    return krw, kro, krg

def stone2(Sw, Sg):
    krw, krg, krow, krog = two_phase(Sw, Sg)
    raw = kro_ow * ((krow / kro_ow + krw) * (krog / kro_ow + krg) - (krw + krg))
    return krw, np.maximum(raw, 0.0), krg

# ---- grid: use meshgrid, then keep the triangle Sw + Sg <= 1 --------------
Sw_1d = np.linspace(Swr, 1.0, 101)
Sg_1d = np.linspace(0.0, 1.0 - Swr, 101)
SW, SG = np.meshgrid(Sw_1d, Sg_1d)
valid = SW + SG <= 1 + 1e-12
Sw, Sg = SW[valid], SG[valid]
So = np.clip(1 - Sw - Sg, 0, 1)
krw, kro1, krg = stone1(Sw, Sg)
_, kro2, _ = stone2(Sw, Sg)
```